61st International Mathematical Olympiad Saint Petersburg Russia IMO 2020

HK IMO Training 2019-2020
Problem Set 18
For 14 March 2020

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Problem 69. Let n be a positive integer. Show that there exists an integer m such that

$$2020m^2 + 20202019m + 2019$$

is divisible by 2^n.

Problem 70. Find the largest positive integer n such that there exist n real polynomials where the sum of any two has no real roots but the sum of any three does.

Problem 71. Choose positive integers $x_1, x_2, \ldots$ satisfying

$$1 = \frac{x_1}{1^2} > \frac{x_2}{2^2} > \frac{x_3}{3^2} > \frac{x_4}{4^2} > \cdots$$

and let α denote the largest real number satisfying

$$\frac{x_n}{n^2} \geqslant \alpha \quad \text{for all positive integers } n.$$

What are the possible values of α over all possible choices of the sequence $\{x_n\}$?

Problem 72. Two circles Γ_1 and Γ_2 have common external tangents ℓ_1 and ℓ_2 meeting at T. Suppose ℓ_1 touches Γ_1 at A and ℓ_2 touches Γ_2 at B. A circle through A and B intersects Γ_1 again at C and Γ_2 again at D, such that quadrilateral $ABCD$ is convex. Suppose lines AC and BD meet at point X, while lines AD and BC meet at point Y. Show that T, X, Y are collinear.